

The Cardiology Inbox Safety Gap

Closing the Loop on Abnormal Results, Imaging Findings, Patient Messages, and Follow-Up Obligations in Small Cardiology Practices

Thesis

Small cardiology practices do not suffer because cardiologists lack judgment. They suffer because modern cardiology generates more follow-up obligations than legacy inbox workflows can safely close.

Prepared for	Pocono AI, LLC
Proposed architecture	Sentinel Node
Primary workflow	Cardiology results-to-closure
Primary audience	Cardiologist-owners, practice managers, physician pilot candidates, investors
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Executive Summary

Small cardiology practices do not suffer because cardiologists lack judgment. They suffer because modern cardiology generates more follow-up obligations than legacy inbox workflows can safely absorb.

A single day in a cardiology office can produce abnormal labs, echocardiogram reports, stress test findings, Holter or Zio monitor reports, medication refill decisions, anticoagulation questions, surgical clearance requests, hospital discharge summaries, outside imaging reports, incidental findings, and patient portal messages. Each item may look small in isolation. Together, they create a high-friction, high-liability operating environment.

This white paper argues that the central problem is not physician productivity in the narrow sense. The central problem is closure: whether the practice can show that each clinically meaningful obligation was routed, reviewed, acted upon, communicated, and documented.

Pocono AI's proposed Sentinel Node is not positioned as an autonomous medical decision-maker. It is positioned as local, human-reviewed workflow infrastructure: a secure on-premise appliance that helps a practice retrieve context, classify inbox items, identify open loops, route work to the right role, draft communications for review, and preserve an audit trail.

The recommended implementation is the Cardiology Results Closure Pilot: a focused pilot for a small cardiology practice with a cardiologist, an APP, nursing support, medical assistants, and a practice manager. The initial pilot should focus on rhythm reports, echocardiogram and stress test results, high-risk labs, selected symptom-bearing portal messages, and incidental findings.

Core finding

The difference between reviewed and closed is where much of the safety and liability exposure lives. Sentinel Node should be designed to make that difference visible.

1. The Problem: Cardiology Creates More Follow-Up Than the Inbox Was Built to Hold

Cardiology is uniquely exposed to inbox overload because it combines chronic disease management, diagnostic testing, urgent symptom triage, medication risk, imaging interpretation, anticoagulation decisions, and procedural clearance. The EHR inbox has become the default intake channel for fragments of clinical obligation, but it was not designed as a complete results-to-closure operating system.

- A potassium result from a heart failure patient on spironolactone.
- A creatinine increase after medication titration.
- An echocardiogram showing reduced ejection fraction.
- A stress test report with an equivocal finding.
- A Holter or Zio monitor report showing nonsustained ventricular tachycardia.
- A patient portal message reporting palpitations, chest pain, dyspnea, dizziness, or watch-detected atrial fibrillation.
- A surgical clearance request requiring anticoagulation guidance.
- A refill request that should not be approved without checking labs.
- A hospital discharge summary changing the medication list or follow-up plan.

- A cardiac CT report that includes an incidental pulmonary nodule.

Each item carries a question: what happens next? The inbox may record that a message or result was opened, but it may not prove that the clinically meaningful loop was closed.

2. The Workflow: Cardiology Results-to-Closure

This white paper focuses on one workflow: the cardiology results-to-closure workflow. It is the practice's ability to move from an incoming clinical signal to a documented, assigned, completed follow-up state.

1. A result, report, or message arrives.
2. The practice identifies what it is.
3. The relevant patient context is retrieved.
4. The correct role is assigned.
5. A clinician reviews where required.
6. The patient, PCP, specialist, or external office is notified where appropriate.
7. Follow-up is ordered, scheduled, transferred, or explicitly documented as unnecessary.
8. The item is closed with an auditable record.

Important distinction

A result can be reviewed but not closed. A patient can be called but not reached. A recommendation can be noted but not scheduled. An incidental finding can be visible in a report but have no documented owner.

3. Pilot Practice Model

The initial pilot should be grounded in a realistic small cardiology setting rather than a hospital-scale implementation. The recommended model is:

- One cardiologist-owner.
- One cardiology APP, such as an NP or PA.
- Two nurses handling triage and clinical calls.
- Two medical assistants handling scheduling, routine calls, records, and follow-up logistics.
- One practice manager responsible for throughput, staffing, bottlenecks, and risk exposure.
- External relationships with hospitals, PCPs, imaging centers, labs, ambulatory rhythm-monitor vendors, surgical offices, and dental offices.

This practice is large enough to have real handoff complexity, but small enough that workflow improvement can be implemented without a hospital-scale transformation team.

4. A Real Day Inside the Cardiology Inbox

7:38 AM - The day begins before clinic begins

The cardiologist opens the EHR before the first patient. The inbox includes 14 lab results, 6 echocardiogram reports, 3 stress test reports, 4 Holter or Zio monitor reports, 9 patient portal messages, 7

refill requests, 5 surgical clearance requests, 3 hospital discharge summaries, 2 outside imaging reports, and 11 staff messages marked "please advise."

The inbox looks like a list. Clinically, it is not a list. It is a pile of obligations. Some items are routine. Some are dangerous. Some are duplicates. Some belong to the APP. Some can be handled by nursing under protocol. Some require the cardiologist. Some require patient notification. Some require follow-up scheduling. Some require ownership transfer. Some are unclear.

8:12 AM - Abnormal lab: potassium and renal function

A lab result returns: potassium 5.7, creatinine elevated from prior baseline, and the patient has heart failure. The medication list includes potassium-affecting therapy.

The clinical decision belongs to the clinician. But before the clinician can decide, the practice has to determine whether this is new, what the trend is, whether the specimen may be hemolyzed, who ordered the lab, whether the lab followed a medication change, whether the patient has symptoms, whether a repeat lab is needed, and who owns the patient call.

9:26 AM - Echo result: reduced ejection fraction

An echocardiogram report returns. The ejection fraction has dropped from approximately 50% to 35%. The result is clinically meaningful, but the report itself does not close the loop. The practice must determine whether follow-up is already scheduled, whether medication optimization needs review, whether symptoms were documented, and whether the patient has been notified.

10:41 AM - Incidental finding on cardiac imaging

A cardiac imaging report includes an incidental pulmonary nodule and recommends follow-up. This is one of the most dangerous categories of workflow failure because ownership is ambiguous. Cardiology ordered or received the study, but the finding is non-cardiac. The PCP may or may not have been copied. The patient may or may not understand the recommendation. The system may not clearly show who owns follow-up.

12:18 PM - Anchor case: Holter/Zio monitor report with rhythm risk

A monitor report returns showing runs of nonsustained ventricular tachycardia. The report may not be a five-alarm emergency in every patient, but it is not a routine inbox item either. It requires clinical context: symptoms, prior echo and ejection fraction, structural heart disease, beta blocker use, prior rhythm history, ischemic evaluation, existing follow-up, and whether the patient has been notified.

Anchor case for the pilot

The monitor report is the ideal pilot anchor because it is cardiology-specific, context-dependent, potentially high-risk, and easy to lose in a generic inbox stream.

2:07 PM - Portal message: "My watch says AFib again"

A patient sends a portal message: "My Apple Watch says I am in afib again. I feel weird and a little short of breath. Should I go to the ER?" This is not a casual message. It may require nurse triage, symptom assessment, medication and anticoagulation context, and urgent escalation under practice-defined rules. Sentinel should not answer this patient directly. The safe workflow is context preparation and routing.

4:36 PM - Surgical clearance and anticoagulation

A dental office or surgical center asks whether the patient can stop Eliquis before a procedure. This appears administrative, but it is not purely administrative. The practice must understand the indication for anticoagulation, stroke risk, procedure bleeding risk, timing, recent symptoms, and whether a clinician must review.

6:18 PM - The second shift

Clinic ends. The inbox remains. The cardiologist reviews reports, signs refills, answers staff questions, looks up prior studies, routes tasks, documents plans, and tries to close loops. This is the point where burnout, risk, and liability merge. The physician is tired, the work is fragmented, and the EHR may record clicks without proving closure.

5. The Core Failure Pattern: Open-Loop Follow-Up

In cardiology inbox management, the most important failure pattern is open-loop follow-up. An open loop exists when an item creates an obligation but the system cannot show that the obligation was completed.

- Abnormal lab reviewed, but no repeat test ordered.
- Echo abnormality acknowledged, but no follow-up appointment scheduled.
- Monitor report reviewed, but patient notification not documented.
- Incidental finding noted, but PCP notification or ownership transfer not documented.
- Portal symptom message routed like a routine message.
- Clearance response completed without documenting the clinical basis.
- Patient called, but no documentation of successful contact or next step.
- Follow-up recommended, but no staff owner assigned.

The EHR may show that something happened. It may not show that the right thing happened. This is why review is not enough. Closure is the standard.

6. Sentinel Node: Local Workflow Infrastructure, Not Autonomous Medicine

Sentinel Node is proposed as a local AI workflow layer for regulated professional environments. For cardiology, its purpose is not to produce autonomous medical decisions. Its purpose is to support human-reviewed workflow closure.

- Intake: collect scoped result and report types into a controlled workflow layer.
- Classification: identify the kind of item and the likely workflow obligation.
- Context retrieval: assemble the patient-specific evidence the clinician needs.
- Risk surfacing: identify open-loop conditions without making final clinical decisions.
- Role routing: move work to cardiologist, APP, nurse, MA, practice manager, PCP, or owner-assignment queue.
- Draft communication: prepare patient, staff, or external-clinician messages for review.
- Closure tracking: maintain visible status until the obligation is closed.

- Audit preservation: retain the source, evidence, human review, action, timestamps, and final status.

7. Intake: Start Narrow, Then Expand

The pilot should not overpromise full enterprise EHR integration on day one. A realistic early implementation can use hybrid intake: controlled exports, secure document upload, PDF reports, selected result files, staff-forwarded packets, FHIR or API integration where available, and deeper HL7/EHR integration later.

The initial scope should include Holter/Zio monitor reports, echocardiogram reports, stress test reports, high-risk labs tied to heart failure and medication safety, imaging reports with incidental findings or follow-up recommendations, selected symptom-bearing portal messages, and clearance requests requiring clinical review.

8. Classification: Separating Noise From Obligation

Sentinel classifies each incoming item into workflow categories. This classification should never be final clinical disposition. It should be a draft workflow label requiring practice-defined review.

- Routine result.
- Abnormal result requiring review.
- Potentially urgent result.
- Monitor/rhythm finding.
- Imaging finding.
- Incidental non-cardiac finding.
- Medication safety issue.
- Anticoagulation question.
- Patient symptom message.
- Surgical/procedural clearance request.
- External result with unclear ownership.
- Duplicate or already-addressed item.
- Needs staff action after clinician review.
- Needs patient notification.
- Needs follow-up appointment.
- Needs repeat lab or repeat study.
- Needs referral.
- Needs closure documentation.

9. Context Retrieval: The Highest-Value Use of AI

The highest-value use of AI in this workflow is not answering clinical questions from general knowledge. It is retrieving the specific patient context the clinician needs. The system does not need to be smarter than the cardiologist. It needs to remove the hunting, assemble the packet, and make open loops visible.

Incoming item	Context Sentinel should	Open loops to check
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	retrieve	
Holter/Zio monitor report	Reason ordered; symptoms; prior rhythm history; recent EKG; last echo/EF; beta blocker or antiarrhythmic use; anticoagulation status when relevant; upcoming appointment.	No clinician review; no patient notification; no follow-up visit; no EP referral decision; no documented plan.
Echocardiogram	Prior echo; EF trend; last cardiology note; heart failure diagnosis; current meds; symptoms; follow-up appointment status.	No patient notification; no appointment; no medication-review plan; no documented no-action rationale.
High-risk lab	Trend; medication context; renal function; ordering reason; last med change; symptoms; repeat lab status.	No repeat lab; no medication review; no patient contact; no clinician owner.
Incidental imaging finding	Ordering clinician; report recommendation; PCP copy status; patient notification; follow-up imaging order; ownership transfer.	No PCP notification; no patient notification; no assigned owner; no tracking date.
Portal symptom message	Relevant diagnoses; meds; anticoagulation; recent test results; urgent symptoms; prior calls; triage protocol.	No triage owner; no escalation; message aging; no documented response.

10. Risk Surfacing Without Autonomous Diagnosis

Sentinel can surface workflow risk without making final clinical determinations. The language must be careful.

Unsafe framing	Safer framing
This patient needs EP referral.	Monitor report includes rhythm finding; clinician review recommended; no follow-up appointment found.
Stop medication.	Medication safety context identified; lab abnormality and medication list retrieved for clinician review.
No urgent action required.	No documented urgent marker found in available sources; clinician review still required.
Patient is safe.	Draft routing packet prepared; final determination remains with clinical team.

11. Role Routing: Moving Work to the Right Human

A cardiology inbox is not only a physician problem. It is a team workflow problem. Sentinel should support routing to cardiologist, APP, RN triage, medical assistant, practice manager, referral coordinator, PCP, external ordering clinician, and a "needs owner assignment" queue.

Most important routing category

Needs Owner Assignment. Ambiguity is a major risk. A system that identifies unclear ownership can reduce the chance that a result disappears because everyone assumed someone else was handling it.

12. Draft Communication

Sentinel may draft patient or clinician-facing communications, but these drafts should require human review. The value is not only speed; it is consistency, documentation, and clear closure.

- Patient notification draft after clinician review: schedule follow-up to discuss monitor findings.
- Repeat lab scheduling draft after clinician sign-off.
- PCP notification draft for incidental finding and ownership confirmation.
- Staff call script draft with practice-approved escalation language.
- Clearance response packet draft for clinician review, not autonomous clearance.

13. Closure Tracking: The Center of the Pilot

Closure tracking is the core deliverable. Each item should have a visible status: new, classified, context retrieved, needs clinician review, reviewed, needs patient notification, patient notified, unable to contact patient, follow-up needed, follow-up scheduled, repeat lab/study ordered, referral placed, PCP notified, ownership transferred, no action required and documented, closed, escalated, or overdue.

The practice should be able to answer one question at any time: What clinically meaningful obligations are still open?

14. Audit Trail: Liability Reduction Through Proof of Process

A safe workflow must produce proof. For each item, Sentinel should preserve the source document, date/time received, classification, retrieved context, evidence links, draft recommendation, assigned owner, human reviewer, final decision, patient communication status, follow-up order or scheduling status, external notification status, closure state, timestamps, and hash/integrity record where appropriate.

The purpose of the audit trail is not defensive medicine alone. It is operational clarity. When a practice can see what happened, who reviewed it, what evidence was available, and whether the loop was closed, the workflow becomes safer.

15. Before and After: The Holter/Zio Anchor Case

Without Sentinel Node

With Sentinel Node

Monitor report arrives as one more inbox item.

Clinician opens report cold between patients or after hours.

Chart hunting is required for symptoms, prior echo, meds, rhythm history, and appointment status.

Follow-up depends on manual routing and memory.

Result may be reviewed but not fully closed.

Audit record is fragmented across EHR clicks, notes, calls, and staff messages.

Monitor report is classified as a rhythm report requiring clinician review.

System assembles a review packet before clinician opens it.

Relevant patient context is retrieved and linked to source evidence.

Draft routing, staff task, and closure checklist are generated for review.

Status remains open until patient notification and follow-up plan are documented or explicitly closed.

Audit trail captures source, context, reviewer, decision, routing, timestamps, and closure state.

16. Cardiology Results Closure Pilot

The pilot should be framed as a workflow-safety and liability-reduction pilot, not as a general AI chatbot deployment.

Pilot promise

Help a small cardiology practice classify, route, document, and close high-risk inbox obligations using a local, human-reviewed AI workflow layer.

Pilot duration

Recommended: 30 to 60 days.

Initial scope

- Holter/Zio/rhythm monitor reports.
- Echocardiogram and stress test results.
- High-risk labs tied to heart failure, renal function, electrolytes, anticoagulation, and medication safety.
- Imaging reports with incidental findings or explicit follow-up recommendations.
- Selected portal messages involving palpitations, dyspnea, chest pain, syncope, anticoagulation questions, or watch-detected rhythm concerns.

Explicit exclusions from the early pilot

- Emergency triage automation.
- Autonomous patient messaging.
- Autonomous medication adjustment.
- Autonomous diagnosis.
- Full EHR replacement.
- Full prior authorization automation.

- All portal messages.
- All practice operations.

17. Pilot Deliverables

9. Current-state workflow map.
10. Role and responsibility matrix.
11. High-risk result taxonomy.
12. Results-to-closure dashboard.
13. Sentinel review packet template.
14. Holter/Zio anchor workflow.
15. Echo/stress result workflow.
16. High-risk lab workflow.
17. Incidental finding ownership workflow.
18. Draft patient communication templates.
19. Draft PCP/external clinician notification templates.
20. Open-loop tracking report.
21. Audit log design.
22. Weekly practice risk summary.
23. Before/after workflow comparison.
24. Final pilot report.

18. Pilot Metrics

- Number of inbound results processed.
- Number of rhythm reports classified.
- Number of abnormal results flagged.
- Number of open loops identified.
- Time from result arrival to classification.
- Time from result arrival to clinician review.
- Time from clinician review to patient notification.
- Time from result arrival to closure.
- Number of items with unclear owner.
- Number of items routed to physician unnecessarily.
- Number of items handled by staff after clinician sign-off.
- Number of patient notification drafts generated.
- Number of follow-up appointments scheduled from flagged results.
- Number of repeat labs/studies ordered from flagged results.
- Number of incidental findings with documented ownership transfer.
- Physician after-hours inbox time estimate.
- Staff "please advise" volume.
- Percentage of high-risk items closed within practice-defined thresholds.
- Physician satisfaction.
- Staff satisfaction.

- Practice manager satisfaction.

19. Why Local Matters

Cardiology inbox data is highly sensitive. It may include diagnoses, medications, symptoms, test results, imaging findings, hospitalizations, procedure plans, anticoagulation decisions, patient messages, and external physician communications. A local architecture is not merely a technical preference. It is part of the trust model.

- Local inference.
- Local retrieval.
- Local audit trail.
- No telemetry.
- No third-party model calls.
- No cloud exfiltration.
- Human-reviewed outputs.
- Evidence-linked summaries.
- Role-based workflow queues.

Positioning sentence

The most sensitive workflow in the practice should not require sending patient context to an external AI service by default.

20. What Sentinel Node Should Not Do

- Diagnose patients autonomously.
- Make final clinical decisions.
- Tell patients they are safe.
- Independently change medications.
- Independently clear patients for surgery.
- Independently determine emergency urgency.
- Replace physician review.
- Replace nursing judgment.
- Replace the EHR.
- Guarantee that malpractice cannot occur.
- Claim to prevent all missed results.

The product should claim workflow support, closure tracking, context retrieval, routing, draft generation, auditability, and local control. It should not claim autonomous clinical authority.

21. Website Proof-Flow Recommendation

The derived website asset should not be a full pick-your-day simulator yet. Version one should be a guided interactive timeline called "A Day Inside the Cardiology Inbox." It should show six cards, each with a side-by-side comparison: Without Sentinel and With Sentinel.

25. 7:38 AM - The inbox starts overloaded.
26. 8:12 AM - High-risk lab needs context.
27. 9:26 AM - Echo result creates follow-up obligation.
28. 10:41 AM - Incidental finding creates ownership ambiguity.
29. 12:18 PM - Holter/Zio report requires context-dependent review.
30. 6:18 PM - The second shift becomes visible and measurable.

This page should prove that Pocono AI understands the physician day in practical detail. It should invite the buyer into a workflow review, not into a gimmick.

22. Go-to-Market CTA

The recommended call to action is: Apply for the Cardiology Results Closure Pilot.

Supporting language: "In 30 to 60 days, Pocono AI will help your cardiology practice map one high-risk results-to-closure workflow, deploy a local human-reviewed Sentinel review layer, and measure whether high-risk inbox obligations become easier to classify, route, document, and close."

23. Conclusion

Cardiology practices do not need another generic AI assistant. They need infrastructure that understands the workflow reality: abnormal results, rhythm reports, imaging findings, patient messages, clearance requests, staff routing, patient notification, follow-up scheduling, and audit trail. The highest-value opportunity is not to replace cardiologist judgment. It is to protect that judgment from being buried inside an overloaded inbox.

Sentinel Node should be positioned as a local, auditable, human-reviewed workflow layer for high-liability professional environments. In cardiology, the first proof point is results-to-closure: making sure clinically meaningful obligations become visible, assigned, reviewed, communicated, and closed.

Final argument

Pocono AI is not selling an AI chatbot to cardiologists. It is selling a local safety layer for the daily workflow where cardiology risk, physician burnout, patient communication, and liability exposure meet.

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